

Lightning common bean

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Smith, T. H., Michaels, T. E., Lindsay, A. M. and Pauls, K. P. 2009. **Lightning common bean**. Can. J. Plant Sci. **89**: 303–305. Lightning is an upright short vine (type II^a) white bean (*Phaseolus vulgaris* L.) cultivar intended for use in areas with greater than 2600 crop heat units. It has excellent yield potential in either wide or narrow row production and is resistant to races 1 and 15 of bean common mosaic virus. Seed has high cooking and canning quality.

Key words: *Phaseolus vulgaris* L., white bean, common bean, cultivar description

Smith, T. H., Michaels, T.E., Lindsay, A. M. and Pauls, K. P. 2009. **Le haricot Lightning**. Can. J. Plant Sci. **89**: 303–305. Lightning est un cultivar nain de haricot blanc (*Phaseolus vulgaris* L.) à type de croissance indéfinie destiné aux régions recevant plus de 2 600 unités thermiques de croissance. Il affiche une excellente productivité en lignes à écartement large ou à écartement étroit et il est résistant aux races 1 et 15 du virus de la mosaïque commune du haricot. Le grain convient pour la cuisson et pour la mise en conserve.

Mots clés: *Phaseolus vulgaris* L., haricot blanc, haricot commun, description de cultivar

The white seeded common bean (*Phaseolus vulgaris* L.) cultivar Lightning (CFIA registration no. 6438) was developed by the Department of Plant Agriculture, University of Guelph, Guelph, Ontario. Lightning has a type II^a plant type with an erect stem and branches and a vine that is shorter in length than that found on OAC Gryphon. The cultivar can be successfully produced in either traditional wide (70 cm) rows or narrower (<40 cm) rows with direct combine harvesting. It is intended for production in areas of Ontario with more than 2600 crop heat units.

Pedigree and Breeding Methods

Lightning was derived from the double cross Navigator/OAC Thunder//AC Compass/OAC Silvercreek made in 1996 in a growth room. Navigator (ROG345) is a navy bean variety from Rogers Seed Company, which was used because of its upright architecture. The pedigree of OAC Thunder (Michaels and Smith 1999) is Ex Rico SL 1351/W84-129 (F3 SPS). AC Compass (Park and Rupert 2000) was derived from Crestwood/HR14//OAC Laser. OAC Silvercreek was derived from OAC Laser/Cran 74Navigator. F₁ plants from the double cross were grown in the field in 1997 at the Elora Research Station, Elora, Ontario, and all seeds were bulked. The F₂ to F₄ generations were advanced using the single pod descent (pod bulk). The F₂ and F₄ generations were grown in a winter nursery in New Zealand. The F₃ generation was grown at the Elora Research Station in the summer of 1998. Single plant selections for early maturity, upright plant habit, and

high pod number were made from F₅ population bulk plots in the field at the Elora Research Station in 1999. The F₅ derived F₆ line (F_{5:6}) was grown in 2000 at the Elora Research Station in the field in plant rows. The F_{5:7} generation was grown in preliminary yield trials in 2001 and the F_{5:8} generation was grown in advanced yield trials in 2002, both at the Elora Research station. A cooperative advanced trial was introduced in 2003 with Dr. S. J. Park at AAFC Harrow. An equal number of F_{5:9} generation entries from the University of Guelph and AAFC were grown in identical yield trials at Elora and St. Thomas. The selection criteria for F_{5:6} to F_{5:9} lines included the same agronomic traits as for the F₅ population as well as high seed yield. Entries from advanced trials were entered into the Ontario Pulse Crop Committee's Registration and Performance trials in 2004 (at St. Thomas, Kippen, Brussels, Elora) and 2005 (at St. Thomas, Kippen, Granton, Elora, and KCAT) as well as grown at the Elora Research Station to produce F_{5:F10} and F_{5:F11} generations. One hundred single plants from the F_{5:11} generation, grown in a growth room in the winter of 2005–2006, were rogued for a uniform plant type and seed from the remaining plants was bulked and sent to Twin Falls, Idaho, USA, for breeder seed production in 2006.

Lightning was tested for resistance to bean common mosaic virus and anthracnose by artificial inoculation in growth rooms at the University of Guelph. Cooking quality was assessed for appearance, flavour and texture by a panel at Agriculture and Agri-Food Canada Greenhouse and Processing Crops Research Centre,

Table 1. Maturity, yield and seed weight of Lightning and three checks in the Ontario Pulse Crop Committee White Bean Variety Registration Trials^z

Cultivar	Yield (kg ha ⁻¹)			Maturity (d)			Seed weight (g 100 seed ⁻¹)
	2004	2005	Mean ^y	2004	2005	Mean ^y	
Lightning	3168	3734	3482	104.8	90.8	97.0	22.7
OAC Gryphon	2846	3929	3447	108.5	96.8	102.0	21.7
OAC Rex	2585	3758	3237	109.5	96.8	102.4	22.3
OAC Thunder	2955	3830	3441	101.5	92.8	96.7	23.4
SE ^x	50	62	59	0.9	0.4	0.6	0.3
No. of trials	4	5	9	4	5	9	9

^zTrials were conducted at St. Thomas, Kippen, Brussels and Elora in 2004 and at Kippen, St. Thomas, Granton, KCATand Elora in 2005.

^yBased on total number of trials.

^xBased on trial means of 31 cultivars.

Harrow, Ontario. Canned bean texture was evaluated using the Ottawa texture measurement system for firmness (N mm⁻¹) and texture (N) (Voisey 1971). A visual examination for the degree of clumping of canned beans, the drain weight of washed-drained beans and the amount of water absorbed by a standard weight of beans were also used to judge cooking quality.

Performance

Seed yield and maturity of Lightning were similar to OAC Thunder (Table 1). Therefore, this variety is appropriate for the short season areas in Ontario greater than 2600 heat units (Brown and Bootsma 1997). The upright plant type (II^a) will facilitate narrow planting and direct combining of the crop.

Lightning has higher cooking quality than both OAC Thunder and OAC Gryphon (Table 2). This is a highly valued trait since most of the white bean production in Ontario is exported and used predominantly for the production of canned foods. Lightning scored 10.0 for

the organoleptic evaluation (visual score of 1 for unacceptably poor and 15 for most acceptable), which was better than the average score (9.2) for two standard canning checks (OAC Thunder and OAC Gryphon) in Ontario in 2005. The hydration coefficient (1.84) was very similar to the check average (1.83). The solid weight (washed-drained weight) of 70.65% was higher than the check mean of 65.58% and much higher than the required weight of 60%. This indicates that the retained solids of canned beans in tomato sauce was very good and acceptable commercially. The texture of canned Lightning beans was acceptable based on firmness values of 23.7 N mm⁻¹ and a plateau force of 292.0 N, which were softer than the average values of the check cultivars (firmness of 28.8 N mm⁻¹ and plateau force of 346.2 N).

Lightning is resistant to races 1 and 15 of bean common mosaic virus, indicating that it carries the I gene, but it is susceptible to races 17 and 23 of anthracnose (Table 2).

Table 2. Cooking quality and disease reaction of Lightning and two checks in the Ontario Pulse Crop Committee White Bean Variety Registration Trials^z

Cultivar	Cooking quality						Disease reaction			
	Panel ^y		Texture ^x	Packing ^w	Drain wt. ^v	Hyd. Coef. ^u	BCMV		Anthracnose	
	(1–15)	Plat. (N)	Firm. (N mm ⁻¹)	(1–5)	(%)		1	15	17	23
Lightning	10.0	292.0	23.70	3.75	70.65	1.84	–	–	+	+
OAC Gryphon	8.9	360.1	28.22	3.25	65.62	1.84	–	–	–	–
OAC Thunder	9.4	338.4	29.37	3.25	65.45	1.82	–	–	+	+
SE ^t	0.106	5.793	0.577	0.100	0.311	0.004				

^zTrials used for cooking quality test were grown at St. Thomas, Granton, Kippen, Elora in 2005.

^yEight panelists evaluated for appearance, flavour and texture of beans canned in tomato sauce using a scale of 1–15; 1, poor; 15, excellent.

^xTexture of washed-drained canned beans measured on Instron Texture measurement system using wire extrusion cells for plateau force (N) and firmness (N mm⁻¹).

^wDegree of clumping of canned beans scored visually from 1 to 5: 1, no clumping; 5, over half clumped.

^vDrain weight, weight of washed-drained canned beans expressed as a percentage of unwashed-undrained weight.

^uSoak weight/dry weight (determined for 500 g of dry beans soaked for soaked for 40 min at 88°C).

^tResistance (–) or susceptibility (+) to bean common mosaic virus (BCMV) races 1 and 15.

^sResistance (–) or susceptibility (+) to Anthracnose races 7 and 23.

^tBased on trial means of 28 cultivars.

Other Characteristics

Lightning has a green hypocotyl in the seedling stage and white flowers, which appear approximately 45 d after planting. At maturity plants are approximately 46 cm in height, taller than both OAC Gryphon (36 cm) and OAC Thunder (40 cm). Pods at maturity are light beige, although slight purple flecking may occur in some environments. Leaves are slightly smaller than those of OAC Thunder and similar in size to those of OAC Rex. Seeds are dull white in colour and similar in size to OAC Rex.

Maintenance and Distribution of Pedigreed Seed

Breeder seed of Lightning is maintained by the Department of Plant Agriculture, University of Guelph, Guelph, Ontario, Canada N1G 2W1. Pedigreed seed is distributed through Hensall District Co-op, P.O. Box 219, Hensall, Ontario, Canada N0M 1X0.

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